

CROP DATA SHEET · SPECIALTY · VINE

Hops

Stronger bines and steadier cones through a hot season.

+8-10%

TYPICAL YIELD GAIN

CROSS-CROP AVERAGE · UP TO ~14% UNDER STRESS

+8-10%

YIELD

Typical gain vs. untreated; greater as heat or drought stress rises.

30-50%

LESS NITROGEN

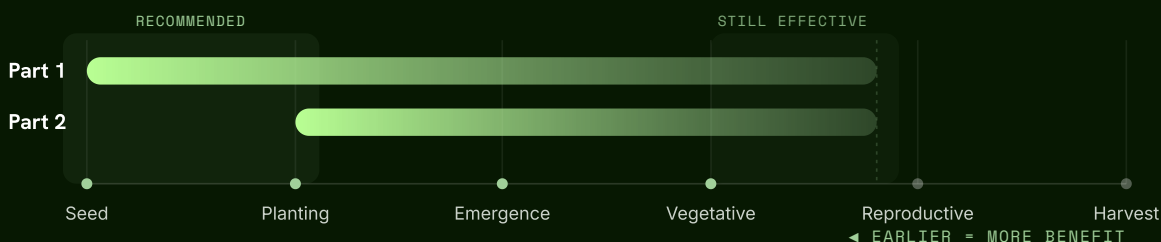
Matched full-rate yield at reduced N (Harman 2011).

Improved

ROOTS & VIGOR

Stronger root development and establishment across trials.

WHEN TO APPLY



Apply by **seed treatment**, **in-furrow** at planting, or as a **foliar pass**. Earlier is better, but a foliar pass well into the **vegetative stage** still delivers a real benefit — it is never too late to start.

HOW THE PROGRAM WORKS

Holds up under stress

Living inputs help the plant manage heat, drought and salinity, so more of the season goes into the crop.

More growth

Symbiotic signalling lifts growth rate and nutrient-use efficiency across the canopy.

Healthier root zone

A richer root environment frees up soil nutrients, so the crop leans less on synthetic NPK.

FIELD-TRIAL EVIDENCE

Hops has not yet been through a Symm-specific field trial, so the figures below are the general cross-crop results from independent peer-reviewed research. They point to where the program helps most in hops: supporting bine vigor and cone development through heat and drought, on less nitrogen.

YEAR	REGION	APPLICATION	RESULT
—	Peer-reviewed (Cornell / Harman)	Trichoderma research	+8-10% yield, improved roots
2011	Peer-reviewed · Harman	Nitrogen study	30-50% N reduction, no yield loss

General cross-crop figures from independent peer-reviewed research, shown as ranges; not hops-specific trial results and never a guarantee for a given field. Yield gain is typically +8 to 10%, up to ~14% under abiotic stress.



See your numbers.

Scan or tap to open our benefit calculator, pre-filled for hops at 100 acres with typical yield and price. Adjust to your farm and see your projected net gain and return.

[ESTIMATE YOUR BENEFIT →](#)

✓ CHEMICAL-FREE

✓ PESTICIDE-FREE

✓ OMRI LISTED

✓ NON-GMO

Build your program for the season.

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